AstroQuantum Realms

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***Abstract* — This paper explores fundamental quantum mechanical phenomena and their implications in astrophysical events, focusing on wave-particle duality, quantum Tunnelling, Decoherence, entanglement, and quantum field theory. It examines how these concepts influence space-time curvature, black hole evaporation, and neutron star mergers. Quantum Tunnelling is analysed in the context of Hawking radiation, highlighting how particles overcome classical energy barriers. The study of Decoherence and entanglement provides insight into quantum information loss in black holes. Quantum field theory is discussed about Hawking radiation and its role in gravitational phenomena. Furthermore, the paper investigates the space-time curvature generated by extreme cosmic events, emphasizing the interplay between quantum mechanics and general relativity. Neutron star mergers are analyzed through gravitational wave signals, revealing their role in heavy element formation and cosmic expansion measurements. Black hole evaporation is explored using Hawking radiation, illustrating mass loss over time. The research highlights how quantum mechanics governs astrophysical phenomena, offering deeper insights into the universe's most extreme environments.**

Keywords— Tunnelling, Decoherence, entanglement, quantum field theory, Hawking radiation

I. INTRODUCTION

Physics, as a fundamental science, is broadly divided into three interrelated domains: classical physics, quantum mechanics, and astrophysics. Classical physics, governed by Newtonian mechanics and Maxwell's equations, describes macroscopic phenomena such as planetary motion, electromagnetism, and thermodynamics. However, it fails to account for the abnormalities observed at atomic and subatomic scales, leading to the development of quantum mechanics. Quantum physics governs the behaviour of particles at microscopic scales, introducing probabilistic interpretations and wave-particle duality[20]. Astrophysics, on the other hand, extends these principles to celestial bodies, exploring the nature of

stars, galaxies, and cosmic structures. Quantum mechanics and astrophysics, two seemingly different fields, intersect in remarkable ways in extreme environments across the universe. These intersections occur in places like black holes, neutron stars, and the cores of stars, where the usual rules of classical physics break down. This intersection remains an open question in theoretical physics, motivating research into quantum field theory in curved spacetime, Hawking radiation, and the possible existence of quantum black holes.

Neutron star mergers, another astrophysical phenomenon, provide a testing ground for these theories. The collision of neutron stars generates extreme conditions of temperature, density, and gravitational fields, pushing the limits of our understanding of nuclear and quantum physics. Such mergers produce gravitational waves, first detected by LIGO(Laser Interferometer Gravitational-Wave Observatory) in 2015, providing direct evidence for Einstein's predictions and giving insights into the behaviour of matter under extreme conditions[21].

Theoretical astrophysics aims to bridge the gap between observations and fundamental physics, employing both classical and quantum principles. String theory and loop quantum gravity are among the major contributors to help in understanding more about quantum mechanics and gravity, with implications that extend to the very fabric of spacetime. ***String Theory*** *proposes that fundamental particles are tiny, vibrating strings, combining quantum mechanics and gravity while suggesting extra dimensions(strings)[22].* ***Loop Quantum Gravity (LQG)****, on the other hand, quantized spacetime itself, depicting it as a network of discrete loops, potentially resolving singularities in black holes and the Big Bang.[23]*

This research explores the intersection of quantum mechanics and astrophysics, focusing on black hole physics, neutron star mergers, and the theoretical advancements that help to understand these domains. The implications of this research extend beyond astrophysics, potentially providing insights into quantum gravity, particle physics, and the early universe.

This project aims to explore these quantum effects in astrophysical contexts. Using Python, we will simulate and analyze quantum phenomena like tunnelling, particle interactions near black holes, and the impact of gravitational fields on quantum states. By bridging the microscopic world of quantum mechanics with the vast scales of astrophysics, this project will shed light on how quantum laws shape energy production, radiation, and particle behaviour in the universe's most extreme environments. Ultimately, this work contributes to our understanding of the cosmos and the fundamental principles that govern it.

II. THEORY

1. WAVE-PARTICLE DUALITY

Wave-particle duality is a quantum mechanical phenomenon describing that particles such as electrons and photons exhibit both wave-like and particle-like properties[24]. The duality was established in Young's double-slit experiment [1]. When particles are not observed going through the slits, they exhibit wave-like behaviour—-spreading out, overlapping, and creating an interference pattern on the screen—like ripples on a pond. When observed, the same particles exhibit particle-like behaviour, colliding at specific points on the screen exhibiting their particle-like behaviour [2].

The duality, mathematically, is represented by the wave function, a solution to the Schrödinger equation, providing a probability distribution for a particle's position or momentum [3]. The wave-like behaviour of particles causes them to

spread out in space and interfere with themselves, essentially contradicting classical notions of localized particles [4].
The concept of wave-particle duality is vividly illustrated by quantum tunnelling, a process in which particles travel potential barriers that laws of classical physics can't cross. This phenomenon demonstrates a fundamental principle of quantum mechanics. The wave-like behaviour of particles causes their probability amplitudes to penetrate and go beyond barriers, resulting in a finite detection probability on the other side [5]. A 2021 study investigated the tunnelling delay time of electrons, emphasising the significance of considering wave and particle properties to understand the tunnelling process fully [5].
These studies demonstrate the intricate interplay between wave-particle duality and quantum tunnelling, developing our understanding of quantum mechanics and its applications.

2. QUANTUM TUNNELING

Quantum tunnelling is a quantum phenomenon where particles traverse energy barriers that they cannot classically surpass. In quantum mechanics, particles are represented by wave functions, which describe their probability of being in different locations[25]. These wave functions can extend into regions that classical physics deems inaccessible. Suppose the energy barrier is sufficiently thin or low. In that case, the particle's wave function gives a probability greater than 0 of appearing on the other side, enabling it to "tunnel" through the barrier.

A helpful analogy is a ball rolling up a hill. In classical physics, the ball must have enough energy to climb over the hill. In quantum mechanics, however, there's a small probability that the ball could bypass the hill entirely, "tunnelling" through it to emerge on the other side—even without the energy required to climb it.

It is used to study how tunnelling applies to **extreme cosmic environments** where classical physics alone cannot explain certain astrophysical processes.

2.1 TUNNELING PROBABILITY

The tunnelling probability quantifies the likelihood that a particle will "tunnel" through an energy barrier it cannot classically overcome[26]. In quantum mechanics, particles are characterized by wave functions. Even when a particle's energy falls below the potential energy of its barrier, the wave function doesn't completely become zero. Rather, it experiences exponential decay in these areas .This non-zero wave function inside and beyond the barrier allows for the possibility of tunnelling.

The probability of a particle tunnelling through a barrier depends on several factors[6]:

i. **Barrier Width and Height:** A thinner or lower energy barrier increases the tunnelling probability.

ii. **Particle Energy:** Higher particle energy relative to the barrier height increases the likelihood.

iii. **Mass of the Particle:** Smaller mass leads to higher tunnelling probabilities since lighter particles have broader wave functions[6].

2.2 MATHEMATICAL EXPRESSION

For a particle facing a potential barrier of height V0 and width L, with the particle's energy E<V0, the tunnelling probability P can be approximated using the **WKB approximation:** $P \propto e^{-2\kappa L}$

$$\kappa = \sqrt{\frac{2m\,(V_0 - E)}{\hbar^2}}$$

where,
κ: Decay constant describing how quickly the wave function diminishes within the barrier
m: Mass of the particle
V_0: Height of the potential barrier
E: Energy of the particle
$\hbar$: Reduced Planck's constant
L: Width of the barrier
This equation shows that tunnelling probability decreases exponentially with:
i. **Barrier Width (L):** as the width increases probability decreases.
ii. **Barrier Height ($V_0 - E$):** A taller barrier also reduces the probability.
iii. **Particle Mass (m):** as the mass increases the probability decreases.

2.3 APPLICATIONS

Quantum tunnelling plays a critical role in astrophysics, especially in extreme environments.

1. Quantum tunnelling allows hydrogen nuclei in stars to combine and form heavier elements, even though they don't have enough energy to overcome their natural repulsion, this is called **Stellar Fusion.**
2. Quantum tunneling explains Hawking radiation by allowing one particle of a virtual pair to escape near a black hole's event horizon, while the other falls in, leading to the black hole's gradual mass loss.

Quantum tunnelling depends on the particle's wave function remaining coherent to pass through a barrier. When a particle interacts with its environment, its wave-like behaviour collapses into a specific state, a process known as decoherence, which lowers the probability of tunnelling[7].

3. DECOHERENCE

Quantum **decoherence** is a critical concept in quantum mechanics, describing the process by which a quantum system loses its coherence[27]. Coherence is the property that allows quantum systems to exhibit superposition and entanglement. This interaction causes the quantum system to "collapse" into a single, definite state, making the system behave more classically.

Decoherence occurs when a quantum system interacts with its environment in such a way that the phase relationships between the components of its wave function become disrupted. This interaction leads to:

- Loss of quantum coherence
- The transition from quantum to classical behaviour
- The apparent collapse of the wave function (though decoherence itself does not explain measurement or "collapse")

Decoherence plays a key role in understanding how quantum information may escape from black holes, particularly regarding the "black hole information paradox." It could provide insight into how information about the quantum states of particles gets lost to the surrounding environment.

4. ENTANGLEMENT

Quantum entanglement is a phenomenon where two or more particles become linked in such a way that the state of one particle instantaneously influences the state of the other(s), no matter the distance between them. This nonlocal behaviour is one of the most intriguing aspects of quantum mechanics, challenging classical intuitions about the separability of physical systems[28]. When particles are entangled, their properties (such as spin, position, or polarization) become correlated, meaning the state of one particle cannot be described independently of the other[8].

This correlation holds even if the particles are separated by vast distances, a property often referred to as **"spooky action at a distance"** by Albert Einstein. Decoherence can disrupt entanglement by causing the system to interact with its surroundings, which destroys the delicate quantum correlations between the particles.

Scientists believe that quantum entanglement plays a role in the phenomenon of Hawking radiation, where pairs of particles become entangled near the event horizon of a black hole. One of the particles falls into the black hole, while the other escapes as radiation, contributing to the gradual evaporation of the black hole.

5. SPACE-TIME CURVATURE RELATION

Space-time curvature *refers to the bending or warping of space and time that happens due to the presence of mass and energy.* According to Einstein's theory of general relativity, massive objects like stars and planets cause space and time to curve around them. This curvature affects how objects move and how time passes[29].

Gravity *is not just a force which pulls objects together, but rather, it is the result of space-time being curved by the mass of an object.* For example, the Earth orbits the Sun because the Sun's mass curves space-time, creating a "path" that the Earth follows. **Time also slows down near very massive objects. The stronger the curvature of space-time, the more time slows down, a phenomenon known as time dilation**[30]**.** In simple terms, space-time curvature is the idea that gravity comes from objects bending space and time, not just pulling on each other.

In regions of strong gravitational fields, such as near black holes, quantum fields must incorporate quantum mechanics and spacetime curvature[12]. For instance, near a black hole's event horizon, wave functions derived from Schrödinger's equation are modified to account for effects like *time dilation* and *spacetime curvature[13]*. In black hole evaporation, it supports quantum field theories that model Hawking radiation and its influence on spacetime. Similarly, neutron star mergers help analyze particle behaviour in extreme gravitational fields and the production of gravitational waves. These adaptations offer insights into the interplay between quantum mechanics and spacetime in the universe's most extreme environments.

6. QUANTUM FIELD THEORY

In quantum physics, Schrodinger's equation is used to relate classical physics to quantum physics. It describes how the energy of the system relates to its wave-like behaviour. It is used to describe the energy of a system at a very small scale(atomic or subatomic level) but fails to recount at such a large scale (like the spacetime plane and such big masses). Since it gives the same results as classical physics for macro-scale objects, we use quantum field theory. It deals with small-scale particles and shows a better relation between large masses and particles.

Quantum Field Theory (QFT) is a framework that combines quantum mechanics and special relativity to describe fundamental forces and particles as quantised fields. In QFT, particles are seen as excitations in their respective fields, and interactions occur through force-carrying fields such as the electromagnetic field (photons) and the gravitational field (hypothetically mediated by gravitons in quantum gravity theories)[9].

QFT is essential for understanding how quantum effects modify macroscopic gravitational phenomena such as black hole evaporation and neutron star mergers. The study of quantum gravity is necessary to resolve singularities and describe the final fate of black holes and the structure of spacetime at the smallest scales[11].

- *7. BLACKHOLE AND NEUTRON-STAR PHYSICS*

7.1 Black-hole

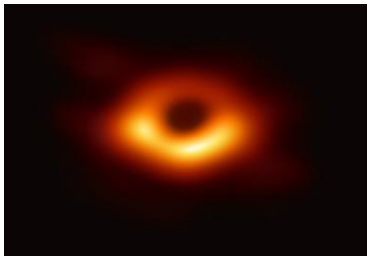
Black-hole is a region of space with gravity so strong that nothing, not even light, can escape from it[14]. It is typically formed when massive stars collapse due to their core gravity at the end of their life cycles.

Key Features:

- *Event Horizon*: refers to the boundary beyond which nothing can escape.
- *Singularity*: A point of infinite density where known physical laws break down.
- *Accretion Disk*: Surrounding material spirals in, heating up and emitting radiation.

Types:

- Stellar mass (few solar masses)
- Supermassive (millions to billions of solar masses, found at galaxy centres)
- Intermediate and primordial (hypothetical types).



[fig: image of the core region of black hole Messier 87] [https://en.wikipedia.org/wiki/Black_hole]

7.1.1 Black-hole Evaporation

Black Hole Evaporation is a theoretical process by which black holes slowly lose mass and energy over time, eventually leading to their disappearance[15]. This idea comes from Hawking radiation, first proposed by physicist Stephen Hawking in 1974, which combines elements of quantum mechanics, general relativity, and thermodynamics.

Here's how it works:

- Close to the point of no return, event horizon (the boundary beyond which nothing, not even light, can escape), virtual particle pairs constantly form and annihilate due to quantum fluctuations.
- Occasionally, one of these particles escapes the black hole while the other falls in[10]. "Hawking radiation" is the term given to the particles that escape from black holes. The temperature of this radiation can be formulated by:

$$T = \frac{\hbar c^3}{8\pi G M k_B}$$

$\hbar$=reduced Planck's constant
c=speed of light
G=gravitational constant
M=mass
k_B= Boltzmann constant

> T (Hawking temperature) increases as mass decreases.

> Small black holes radiate more and evaporate faster.

- This phenomenon leads to a gradual decrease in the energy and mass of the black hole, causing it to shrink and ultimately disappear. This time can be formulated as:

$$t_{evap} = \frac{5120\pi G^2 M^3}{\hbar c^4}$$

ħ=reduced Planck's constant
c=speed of light
G=gravitational constant
M=mass

> This equation gives the total time a black hole takes to evaporate.

> Smaller black holes evaporate much faster than larger ones.

- As time passes, the black hole's mass continues to reduce. When M→0, the black hole disappears.

$$M(t) = M \times (1 - \frac{t}{t_{evap}})^{\frac{1}{3}}$$

- Hawking Radiation Power (Stefan-Boltzmann Law for Black Holes) is given by:

$$P = \frac{\hbar c^4}{15360\pi G^2 M^2}$$

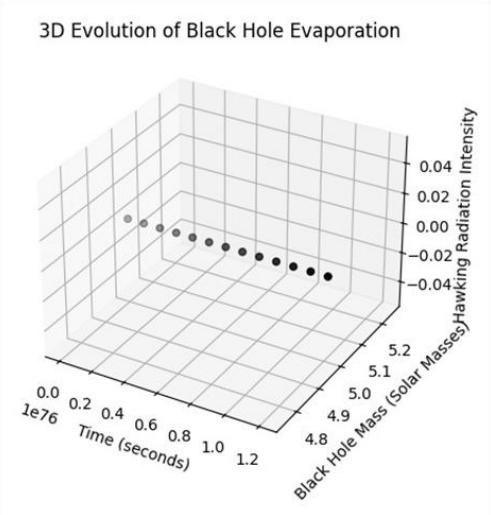
where,

P is the power radiated (Watts)
ħ is the reduced Planck's constant
c is the speed of light
G is the gravitational constant
M is the mass of the black hole
Using the various equations mentioned above, we calculate Hawking's temperature and loss of mass of black holes over time. This can be numerically shown as follows:

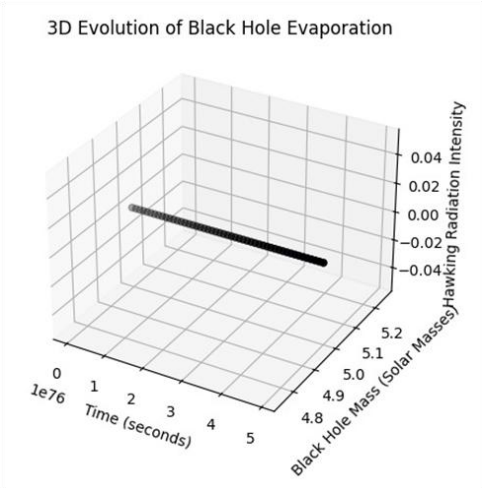
[Table 1: Mass loss due to release of Hawking Radiation overtime]

**Considering the mass of the Black Hole is 5 times the solar mass(i.e. 1.989e30 kg)*

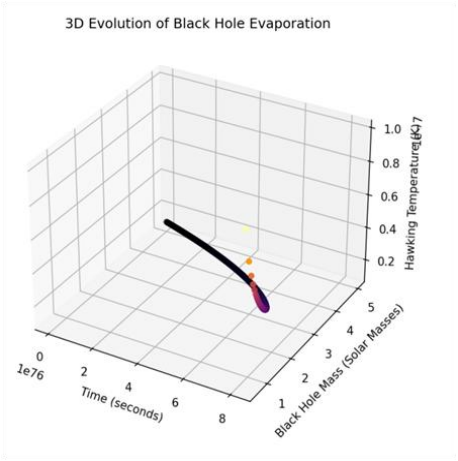
Scaled Time	Time (s)	Mass (Solar Masses)*	Temperature (K)
0.0	0.0	5.0	1.236202e-08
0.062124	5.125619e+75	4.894238	1.262916e-08
0.501002	4.133564e+76	3.96585	1.558559e-08
0.541082	4.464249e+76	3.856692	1.602671e-08
0.995992	8.217525e+76	0.79423	7.782390e-08
0.997996	8.234060e+76	0.630381	9.805197e-08



fig(a) : 1st stage of Blackhole Evaporation



fig(b) : 2nd stage of black hole evaporation



fig(c): Final stage of black hole evaporation

The graph above illustrates a gradual decrease in the mass of a black hole over time due to the emission of Hawking radiation. The following observations can be made based on this :

- **Mass decreases over time:** A black hole steadily loses mass as it radiates energy in the form of radiation. Initially, the decrease is slow, but as the mass reduces, evaporation accelerates.

- **Exponential decay pattern:** A black hole follows an exponential decay pattern, i.e., small black holes evaporate more rapidly than the larger ones. This aligns with Hawking's theory, which states that black holes with smaller masses radiate energy more intensely.

- **Final evaporation stage:** As the black hole approaches the zero-mass stage, its evaporation becomes extremely rapid. This suggests that microscopic black holes could release a final burst of energy before disappearing completely.

This phenomenon highlights the importance of quantum effects in astrophysics, illustrating the interaction between quantum field theory and curved spacetime near the event horizon. Hawking radiation bridges the gap between **general relativity and quantum mechanics.** The numerical data previously available matches the graphical representation hence, verifying the theory of black hole evaporation[ref. Table1]

7.2 Neutron Stars

Neutron stars are extremely dense remnants of supernova explosions, composed mostly of neutrons. It is a result of the collapse of a massive star's core, where protons and electrons combine to form neutrons[16].

Key Features:

- **Density**: around 4×10-17 kg/m³, comparable to nuclear density, where matter is compressed so tightly that atomic nuclei merge, forming a degenerate neutron-rich core.

- **Size**: About 10–20 km in diameter.

- **Magnetic Field**: one of the strongest in the universe.

- **Spin**: Often rotates rapidly, emitting beams of radiation as *pulsars* (a rapidly spinning neutron star that emits radio waves in pulses).

7.2.1 Stellar Fusion

Stellar fusion and neutron star physics are closely linked as the life cycle of stars drives both phenomena. Stellar fusion powers stars by converting hydrogen into helium and heavier elements, releasing vast amounts of energy. When fusion in massive stars stops, they collapse under gravity, forming neutron stars—dense remnants where quantum effects like neutron degeneracy pressure dominate, shaping their extreme physical properties[17].

Stellar fusion refers to the process by which stars generate energy through nuclear reactions in their cores. The energy released in these processes powers stars and produces the light and heat we observe from Earth.

7.2.2 Neutron Star Mergers – A Cosmic Collision

*A **neutron star merger** happens when two neutron stars crash into each other.* This is one of the most powerful events in the universe, producing **gravitational waves (ripples in space-time)**, bright flashes of light, and even heavy elements like gold and platinum.

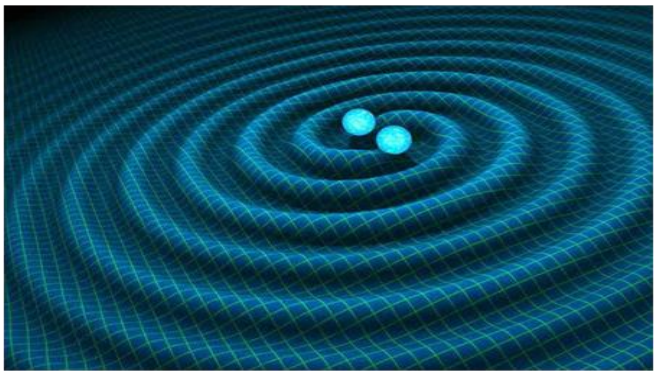


Fig: Two neutron stars influencing each other's gravity [https://aasnova.org/wp-content/uploads/2017/08/fig1-5.jpg]

7.2.2.1 How Do Neutron Stars Merge?

1. **A Pair of Neutron Stars Forms**: If two stars are close together, influencing each other's gravitational field, they can form *a binary neutron star system*[18], orbiting each other like a dance in space.
2. **Getting Closer Over Time**: The two neutron stars slowly move closer because they give off gravitational waves, which take energy away from their orbit. This process can take millions of years.
3. **The Final Crash**: When the neutron stars finally collide, they create[19]:

- **A kilonova**: a massive explosion brighter than a billion Suns.
- A potential **black hole** may form, based on the total mass of the two merging neutron stars.
- A **gamma-ray burst**, a super-powerful blast of energy.

The gravitational wave strain h(t) follows:

$$h(t) \propto \frac{GM}{Rc^3}$$

where ***M*** is the total mass and R is the separation distance.

7.2.2.2 Einstein's Equation

When two neutron stars collide, the event is governed by **Einstein's Field Equations (EFE)**, which describe how matter and energy warp spacetime. the full nonlinear equations include:

The full non-linear equation is:

$$G_{\mu v} = \frac{8\pi G M^3 T_{\mu v}}{c^4}$$

where:

$G_{\mu v}$ is the Einstein Tensor (describes spacetime curvature)
$T_{\mu v}$ is the Stress-energy Tensor (describes matter and energy)
G is the gravitational constant
c is the speed of light

7.2.2.3 Numerical Calculations of Space-Time Curvature during Neutron Star Collision

This prediction models the spacetime curvature during a neutron star collision using the following key equations:

1. Distance to Each Neutron Star:

$$r_1 = \sqrt{(\frac{X + d}{2})^2 + Y^2}$$

$$r_2 = \sqrt{(\frac{X - d}{2})^2 + Y^2}$$

where,

- r_1 and r_2 are the distances from each neutron star

- d is the distance between the two stars, which decreases over time

2. Schwarzschild-like Spacetime Curvature Approximation:

$$warp = -\left(\frac{Gm_1}{r_1 + 0.5} + \frac{Gm_2}{r_2 + 0.5}\right)$$

3. Gravitational Waves Contribution:

$$waves = 0.15 \times \frac{sin(2\pi(r_1 + r_2 - ct)/1.5)}{1 + r_1 + r_2}$$

4. Neutron Star Centers:

$$star_1 = -1.5 \times e^{-5r_1}, star_2 = -1.5 \times e^{-5r_2}$$

5. Total Spacetime Curvature:

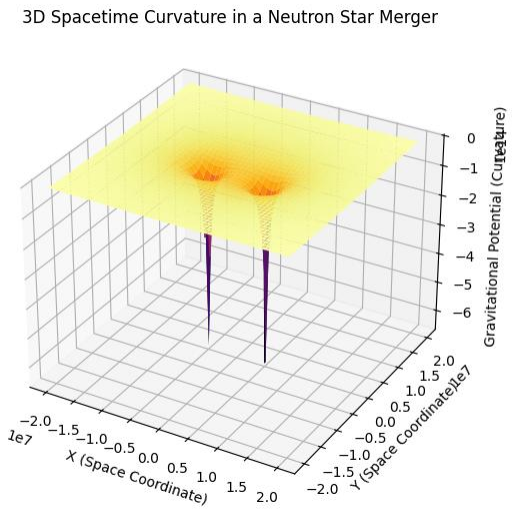
$$Z = warp + wave + star_1 + star_2$$

Using these equations, we calculate space-time warp during neutron-star mergers. The numerical data obtained is as follows:

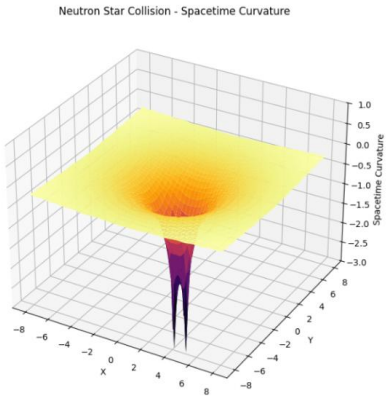
[Table 2: Decrease in separation of two neutron stars influencing each other's gravitational fields over time]

Frame	Time (scaled sec)	Distance (in light years)	Max Curvature	Min Curvature
0	0.0	6.0*	-0.23516	-3.998543
15	1.5	5.28	-0.233547	-3.979018
125	12.5	0.5	-0.237834	-5.196212
140	14.0	0.5	-0.237834	-5.196212
107	10.7	0.864	-0.2387778444229092	-4.8306165724923
124	12.4	0.048	-0.2388425030655518	-7.3668406999039 5

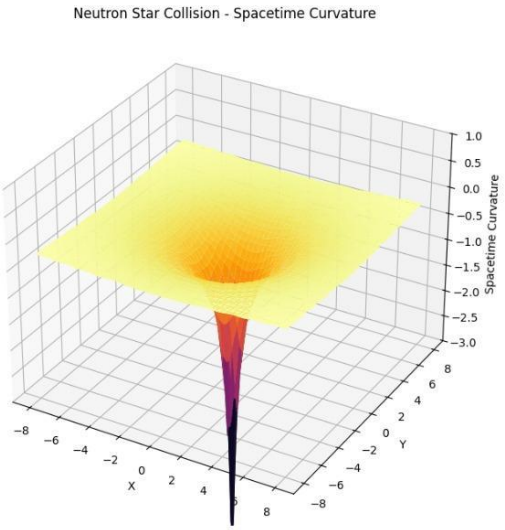
** 6 is taken as the initial distance, if separation is too large (d>10) then inspiral takes too long and the stars would not influence each other's gravitational fields. According to Schwarzschild Radius (i.e. d=4), will be too close and the merger process will be speedy. Hence d=6 is the suitable initial distance.*



fig(a): Before neutron star merger



fig(b): During Neutron Star Merger



fig(c): After Neutron Star Merger

From the above graphs, we observe the following:

- Fig.(a) illustrates two distinct gravitational wells (spacetime curvatures) formed due to the presence of two neutron stars. The two curvatures are relatively symmetric and maintain separate identities.

- Fig.(b) illustrates how the gravitational field of one neutron star affects that of the other, causing their spacetime curvatures to combine, creating a more pronounced gravitational well.

- Fig.(c) illustrates the later stage of the merger, where the neutron stars have fully collided, forming a single, deeper spacetime curvature.

- The transition from two wells to a single, more intense curvature highlights the dramatic effect of a neutron star collision on spacetime.

- This indicates an extreme warping of spacetime, potentially resulting in the creation of either a supermassive neutron star or a black hole.

From *fig. (a), (b), (c)* and *Table (2)* we can conclude that the graphs and the theory match and numerical data is verified.

7.2.2.2 Effects of Neutron Star Merger

1. These collisions make elements like gold, platinum, and uranium, which later spread into space and form planets.

2. They Send Out Gravitational Waves. In 2017, scientists detected gravitational waves from a neutron star merger (GW170817) for the first time[31]. This discovery confirmed Einstein's General Relativity and helped us better understand the universe.

3. By studying these mergers, scientists can measure how fast the universe is expanding.

○

8. ENERGY GENERATION IN BLACK HOLE EVAPORATION AND NEUTRON STAR COLLISION

Energy generation in stellar fusion and black hole evaporation arises from quantum phenomena. In a neutron star merger, an enormous amount of energy (up to 10^{47} joules, equivalent to thousands of supernovae) is released through gravitational waves, gamma-ray bursts, and kilonova explosions. In black hole evaporation, quantum processes like Hawking radiation convert the mass of the black hole into energy, slowly radiating it away into space. Both processes demonstrate how quantum mechanics governs energy transformation in the cosmos.

Gravitational Wave Amplitude Function models a decaying sinusoidal wave, representing how the amplitude of the gravitational wave changes over time. The formula used is:

$$h(t) = e^{-0.2\cdot|t|\cdot sin(10\cdot t)}$$

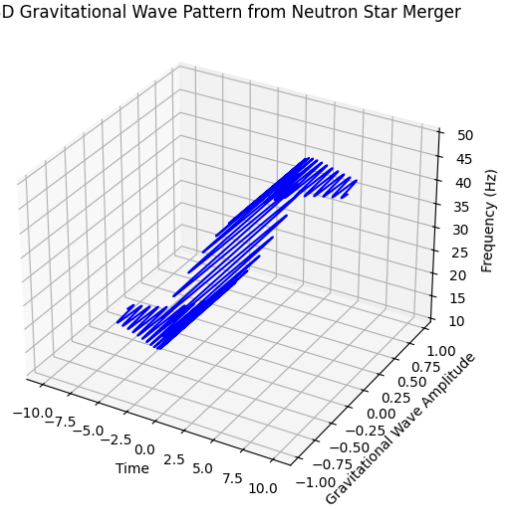
where,

- $h(t)$ is the amplitude at time t
- $e^{-0.2\cdot|t|}$ is the exponential decay factor (causes amplitude to decrease as time moves away from zero)
- $sin(10 \cdot t)$ is the sinusoidal oscillation with a frequency of 10 radians per second

Using the given equation a data set was formed, providing the following numerical values:

[Table 3: numerical data representing the fluctuation in time, amplitude and frequency during neutron star merger]

TIME (seconds)	AMPLITUDE	FREQUENCY(Hz)
-10	0.6853	10.0508
-9.01902	-0.13061	10.0977
-0.01001	-0.09973	29.9333
0.13013	0.93914	30.867
9.01902	0.13061	49.9023
10	-0.06853	49.9492



[fig: graphical data representing the fluctuation in time, amplitude and frequency due to neutron star merger]

The graph illustrates the gravitational wave amplitude over time during a neutron star merger. The following detailed observations can be drawn from the above graph:

- **Gravitational Waveform Evolution:** The intensity of the waveform increases as the stars move toward each other, reaching their peak near the moment of collision.
- **Amplitude Growth:** Initially, the wave amplitude is relatively small, indicating weak gravitational waves. As the merger progresses, the intensity of gravitational interaction increases, thereby increasing the amplitude.
- **Post-Merger Signal:** After reaching its peak, the waveform stabilizes, indicating the formation of a single

dense object, such as a supermassive neutron star or a black hole, which emits waves at a comparatively lower amplitude.

The graph effectively depicts the progression of gravitational waves in terms of time and amplitude, verifying the numerical data and showcasing the crucial stages of approach, merger, and ringdown—aligning with the theoretical predictions of general relativity.

III. Critical Observation

The study examines how extreme astrophysical events, such as black hole evaporation and neutron star collisions, affect space-time curvature (gravitational field). The following are the key observations from the analysis:

1. Distortion in space-time due to neutron star collision:

The gravitational field of two neutron stars influences each other, intensifying the spiral *inwards*. The graph represents two individual gravitational wells merging into one highly intensified well, resulting in extreme space-time curvature. The collision produces intense post-merger signals, frequently resulting in the formation of a single massive neutron star or black hole.

2. Space-time evolution due to black hole evaporation:

It was noted that as the energy and mass of the black hole diminish, the Hawking radiation intensifies. As this process continues, the event horizon contracts and the spacetime curvature surrounding it becomes weak. The graphical analysis of Hawking Radiation shows an *exponential decay* in blackhole mass over time, accelerating as the blackhole reaches its final stage. In this stage, an energy burst occurs, potentially altering space-time in its surroundings.

3. Energy transformation and gravitational waves:

Both these events (i.e. black hole evaporation & neutron star collision) release an immense amount of energy in the form of gravitational waves (ripples) altering the surrounding spacetime. The graphical representation of gravitational waves highlights the increase in amplitude as neutron stars approach the merger and stabilize in the post-collision phase. In black holes, as evaporation nears its end, an energy burst occurs, intensifying space-time curvature and potentially altering the gravitational field surrounding it.

4. Verification of graphical discoveries:

Based on the available datasets and generated graphs, it can be concluded that the theoretical predictions regarding spacetime distortions caused by extreme cosmic phenomena, such as black hole evaporation and neutron star mergers, are potentially verifiable.

IV. Conclusion

This research highlights the significant impact of neutron star collisions and black hole evaporation on space-time curvature (gravitational fields). The special feature of this paper is its graphical representations, showing the distortion in space-time due to above mentioned phenomenon. The analysis indicates that:

- Neutron star mergers intensify gravitational curvature, leading to kilonovas, gamma-ray bursts, extreme tidal forces, and potential black hole formation.
- Black hole evaporation, governed by Hawking radiation, gradually weakens space-time curvature until the black hole disappears, redistributing energy back into the universe.
- Both phenomena bridge quantum mechanics and general relativity, providing key information on the interrelation between energy, gravity, and space-time structure.

The graphical elements, which were previously less prominent, reinforce the theoretical findings and provide clear visualisations of space-time curvature changes throughout these astrophysical processes. These observations help improve our understanding of gravitational wave physics and the quantum nature of black holes, guiding further research in quantum gravity and astrophysical modelling.

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